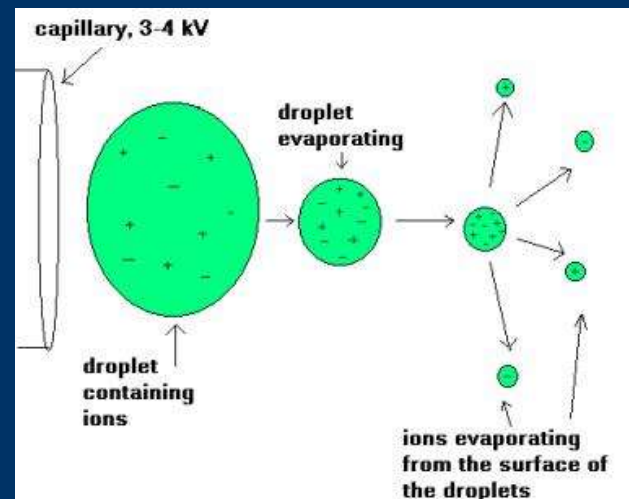
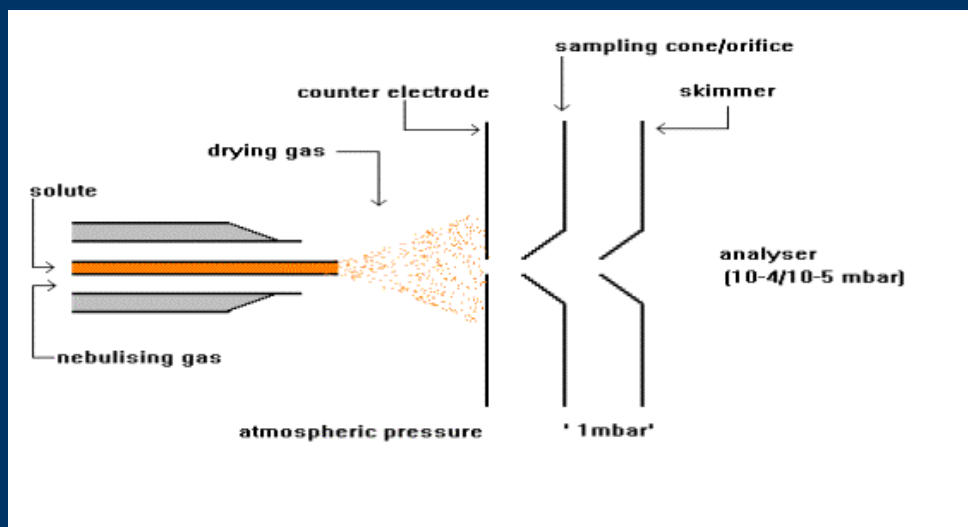


Electrospray ionisation

- **Electrospray Ionisation (ESI)** is one of the **Atmospheric Pressure Ionisation (API)** techniques and is well-suited to the analysis of polar molecules ranging from less than 100 Da to more than 1,000,000 Da in molecular mass.

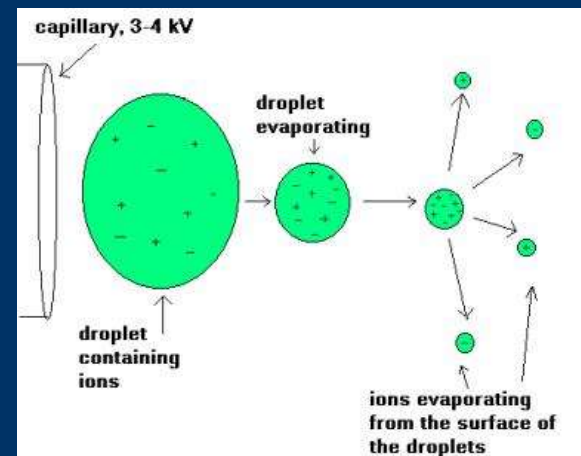
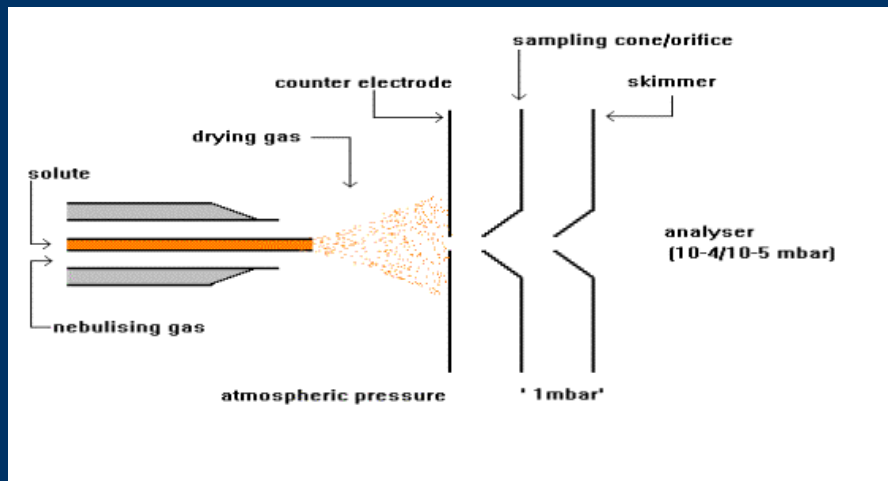
- It has undergone remarkable growth in recent years and is frequently used for LC/MS of thermally labile and high molecular weight compounds. The electrospray is created by applying a large potential between the metal inlet needle and the first skimmer in an API source (Figure). The mechanism for the ionization process is not well understood and there are several different theories that explain this ionization process. One theory is that as the liquid leaves the nozzle, the electric field induces a net charge on the small droplets. As the solvent evaporates, the droplet shrinks and the charge density at the surface of the droplet increases. The droplet finally reaches a point where the coulombic repulsion from this electric charge is greater than the surface tension holding it together. This causes the droplet to explode and produce multiply charged analyte ions. A typical ESI spectrum shows a distribution of molecular ions with different charge numbers.



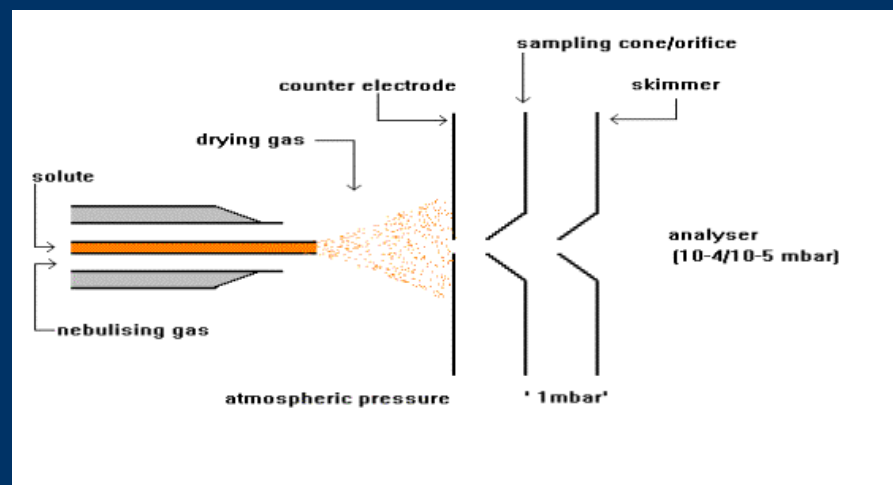
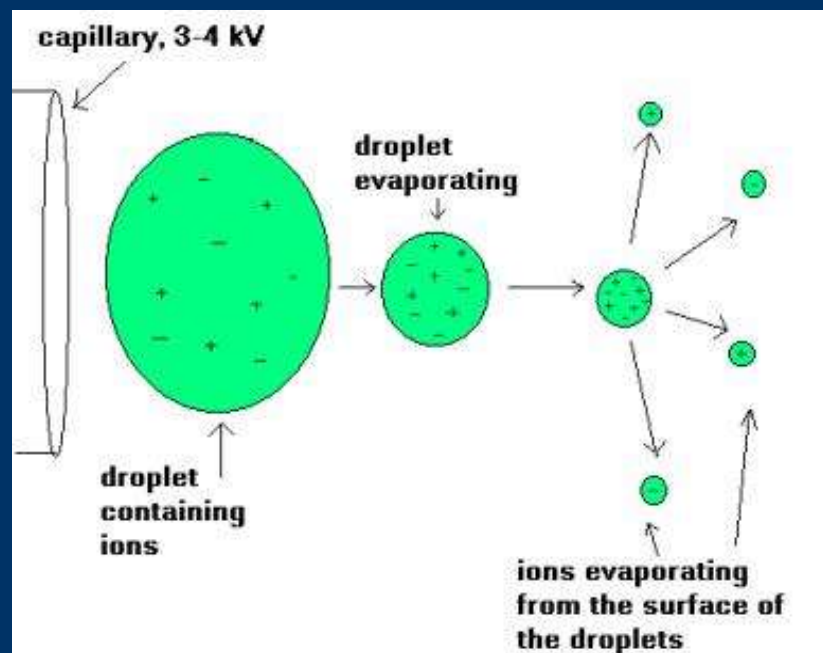
- Because electrospray produces multiply charged ions, high molecular weight compounds are observed at lower m/z value. This increases the mass range of the analyzer so that higher molecular weight compounds may be analyzed. An ion with a mass of 5000 u and a charge of +10 is observed at m/z 500 and is easily analyzed.

Standard electrospray ionisation source

- During standard electrospray ionisation the sample is dissolved in a polar, volatile solvent and pumped through a narrow, **stainless steel capillary** (75 - 150 micrometers i.d.) at a flow rate of between 1 μ L/min and 10 μ L/min. A **high voltage** of 3 or 4 kV is applied to the tip of the capillary, which is situated within the ionisation source of the mass spectrometer, and as a consequence of this strong electric field, the sample emerging from the tip is dispersed into an **aerosol of highly charged droplets**, a process that is aided by a co-axially introduced **nebulising gas** flowing around the outside of the capillary. (REF: J. Fenn, J. Phys. Chem., 1984, 88, 4451).



This gas, usually nitrogen, helps to direct the spray emerging from the capillary tip towards the mass spectrometer. The charged droplets diminish in size by **solvent evaporation**, assisted by a warm flow of nitrogen known as the **drying gas** which passes across the front of the ionisation source. Eventually charged **sample ions**, free from solvent, are released from the droplets, some of which pass through a **sampling cone** or orifice into an **intermediate vacuum** region, and from there through a small aperture into the analyser of the mass spectrometer, which is held under **high vacuum**.



Electrospray Ionization (ESI)

Used to analyze polar, neutral and ionic molecules

Natural products, drug and drug metabolites, proteins, nucleic acids etc.

Mol. Wt. range 100,000 and higher

Can be directly coupled with HPLC to analyze complicated mixtures

Environmental samples, drug metabolites, protein digests etc.

Often produces multiply charged ions

High MW compounds can be analyzed on low m/z analyzers

Easily automated to run samples 24 hrs/day

One person can “operate” multiple instruments

High throughput allows running 1 sample per minute

No fragmentation

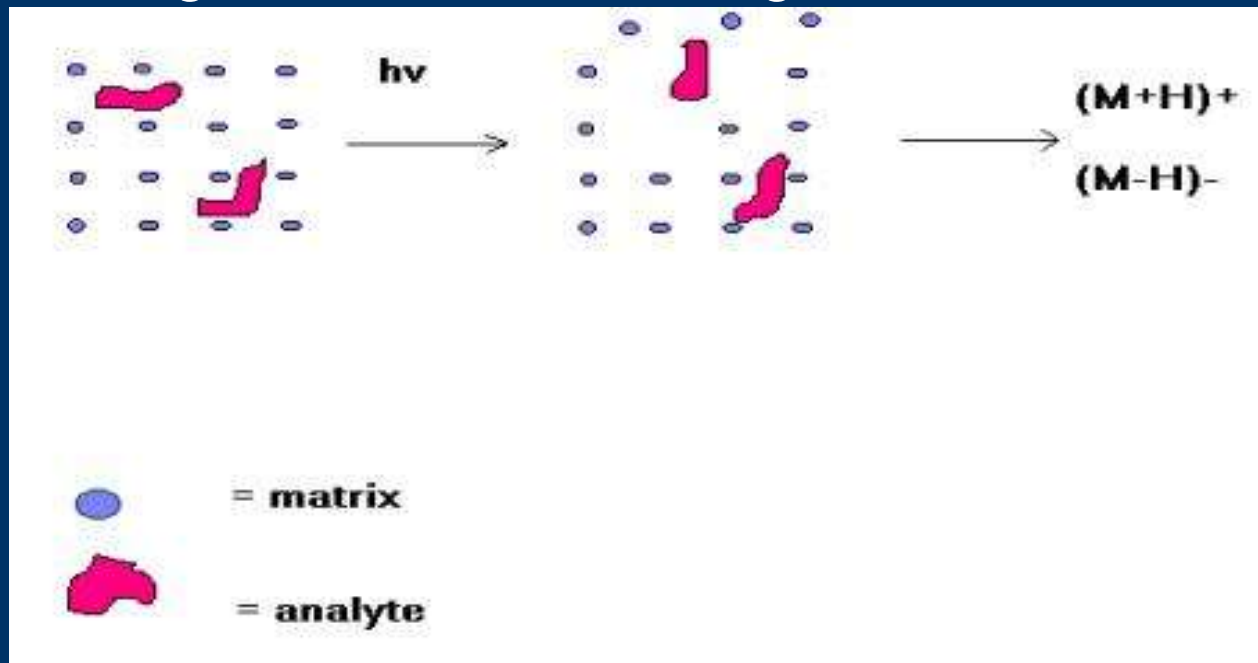
Most frequently used ionization method

Courtesy: Dr. David Smith

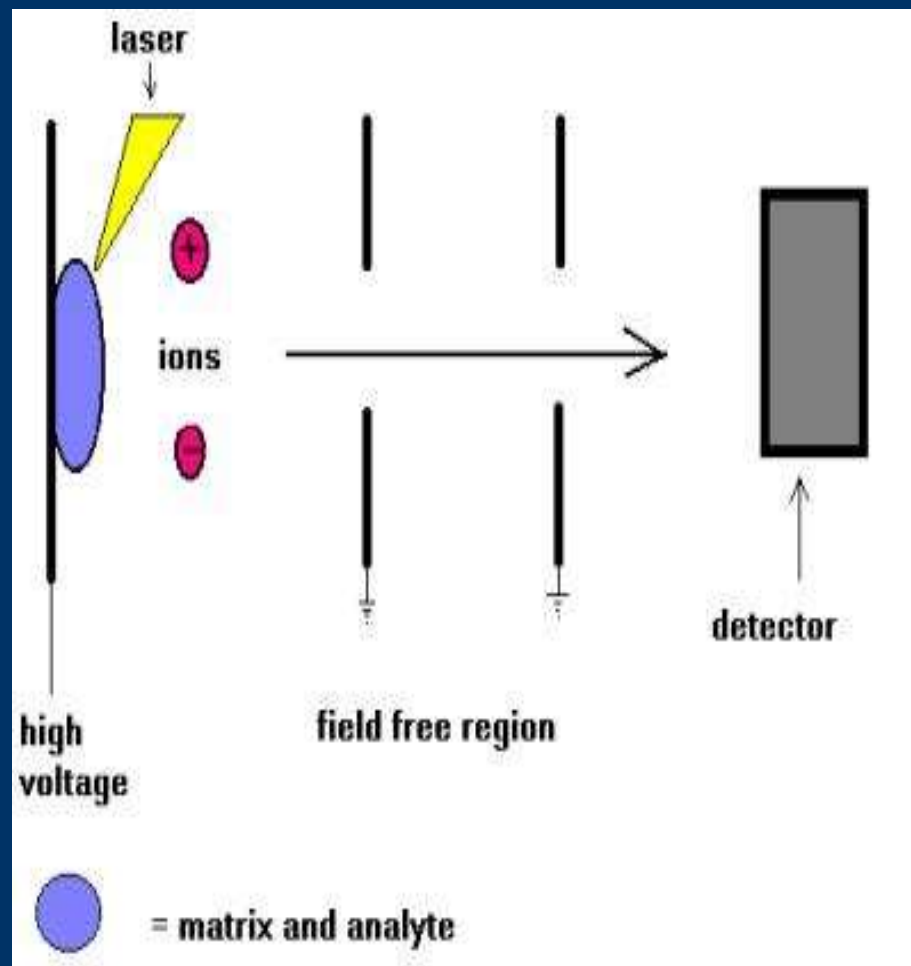
Matrix assisted laser desorption ionisation

- **Matrix Assisted Laser Desorption Ionisation (MALDI)** deals well with thermolabile, non-volatile organic compounds especially those of high molecular mass and is used successfully in biochemical areas for the analysis of **proteins, peptides, glycoproteins, oligosaccharides, and oligonucleotides**. It is relatively straightforward to use and reasonably tolerant to buffers and other additives. The mass accuracy depends on the type and performance of the analyser of the mass spectrometer, but most modern instruments should be capable of measuring masses to within 0.01% of the molecular mass of the sample, at least up to ca. 40,000 Da.
- (REF: F. Hillenkamp, M. Karas, R. C. Beavis, B. T. Chait, *Anal. Chem.*, 1991, **63**, 1193)

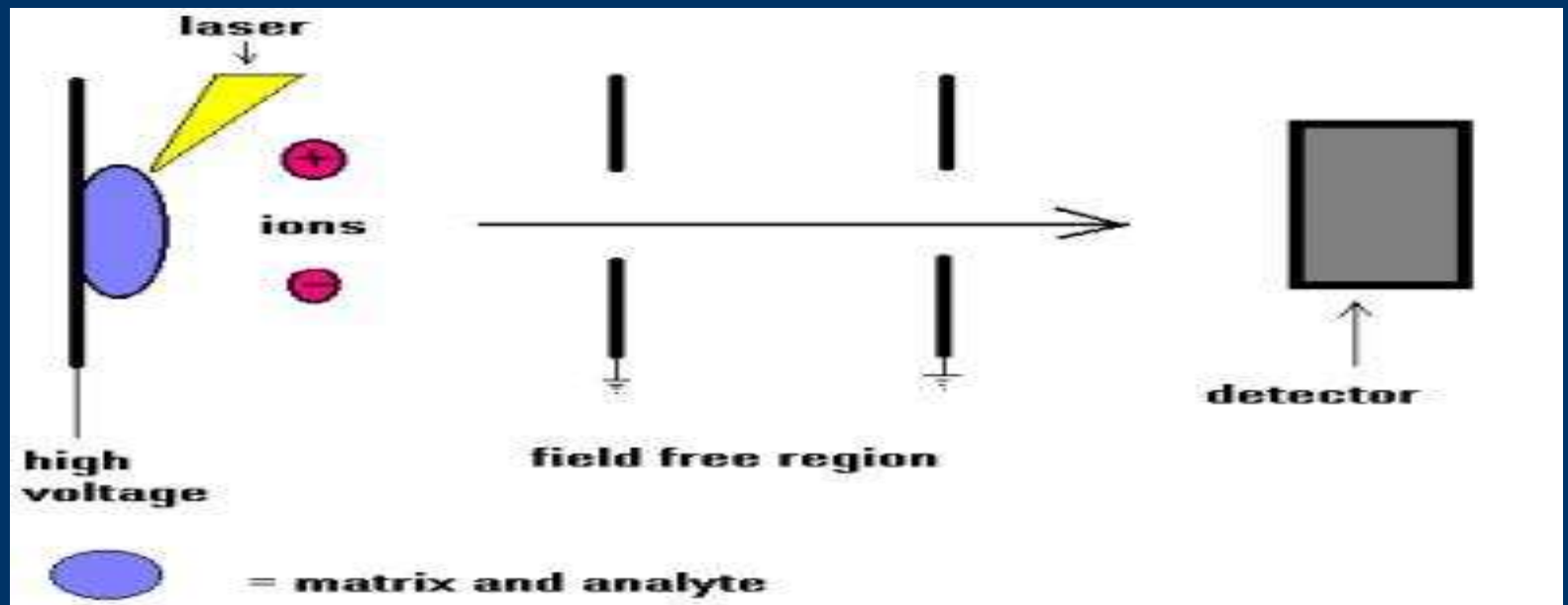
MALDI is based on the **bombardment** of sample molecules with a **laser** light to bring about **sample ionisation**. The sample is pre-mixed with a highly absorbing **matrix** compound for the most consistent and reliable results, and a low concentration of sample to matrix works best. The matrix transforms the laser energy into **excitation energy** for the sample, which leads to sputtering of analyte and matrix ions from the surface of the mixture. In this way energy transfer is efficient and also the analyte molecules are spared excessive direct energy that may otherwise cause decomposition. Most commercially available MALDI mass spectrometers now have a pulsed nitrogen laser of wavelength 337 nm.



- The sample to be analysed is dissolved in an appropriate volatile solvent, usually with a trace of trifluoroacetic acid if positive ionisation is being used, at a concentration of ca. **10 pmol/ μL** and an aliquot (**1-2 μL**) of this is removed and mixed with an equal volume of a solution containing a vast excess of a matrix. A range of compounds is suitable for use as matrices: **sinapinic acid** is a common one for **protein** analysis while **alpha-cyano-4-hydroxycinnamic acid** is often used for **peptide** analysis. An aliquot (1-2 μL) of the final solution is applied to the sample target which is allowed to dry prior to insertion into the high vacuum of the mass spectrometer.



The laser is fired, the energy arriving at the sample/matrix surface optimized, and data accumulated until a m/z spectrum of reasonable intensity has been amassed. The time-of-flight analyser separates ions according to their **mass(m)-to-charge(z) (m/z)** ratios by measuring the time it takes for ions to travel through a field free region known as the flight, or drift, tube. The heavier ions are slower than the lighter ones.



Positive or negative ionisation?

- If the sample has functional groups that readily accept a proton (H^+) then positive ion detection is used **e.g. amines $\text{R-NH}_2 + \text{H}^+ = \text{R-NH}_3^+$ as in proteins or peptides.**
- If the sample has functional groups that readily lose a proton then negative ion detection is used **e.g. carboxylic acids $\text{R-CO}_2\text{H} = \text{R-CO}_2^-$ and alcohols $\text{R-OH} = \text{R-O}^-$ as in saccharides or oligonucleotides**

- In **negative ionisation** mode the **deprotonated molecular ions $(M-H)^-$** are usually the most abundant species, accompanied by some salt adducts and possibly traces of dimeric or doubly charged materials. Negative ionisation can be used for the analysis of **oligonucleotides** and **oligosaccharides**.

SOFT IONIZATION TECHNIQUES

- These techniques attempt to overcome the main drawbacks of EI, namely.
- Absence of M^+ in many cases and hence Mol.Wt. and Mol.Formula.
- The presence of rearrangement reactions.
- The possibility of pyrolysis during vapn. Particularly of less volatile substances.

4.7 FD spectra

FD gives spectra that are normally dominated by M^+ and/or MH^+ . It is very useful for the molecular weight determination of relatively small, highly polar molecules (e.g. D-glucose, Fig. 4.30) or of molecules in the molecular weight range 500–3000 Daltons of intermediate polarity (i.e. polar and yet not sufficiently so to be readily soluble in glycerol for study by FAB or SIMS).

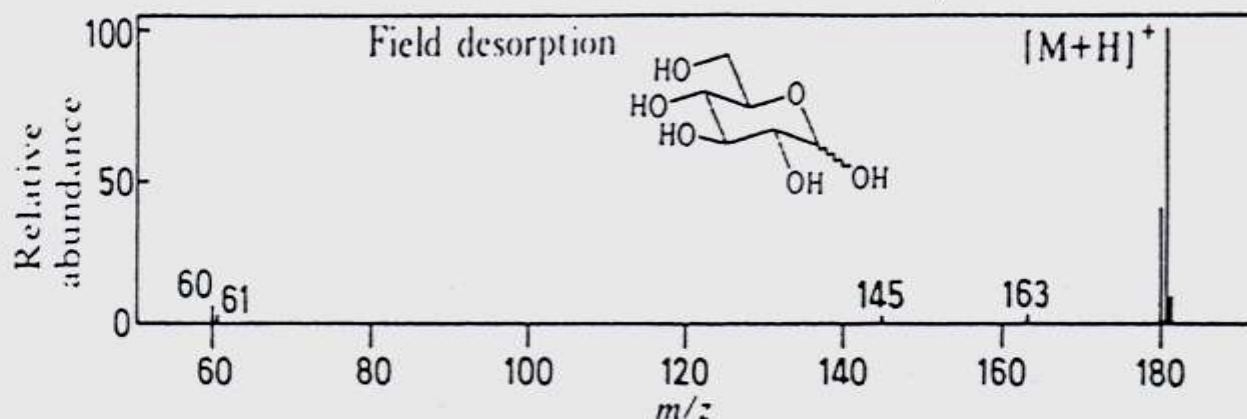


Fig. 4.30

(Reprinted with permission from Howe, Williams, and Bowen, 1981.)

4.8 FAB spectra

Salts (A^+B^-) and highly polar molecules of molecular weights up to *ca.* 3000 Daltons are analysed well by this technique. The former point is illustrated by the positive-ion spectrum (Fig. 4.31) of a phosphonium bromide where A^+ (m/z 349) is the most abundant ion. In contrast, hydrocarbons and other highly hydrophobic molecules (e.g. steroids having little functionality) are not handled well by particle bombardment methods.

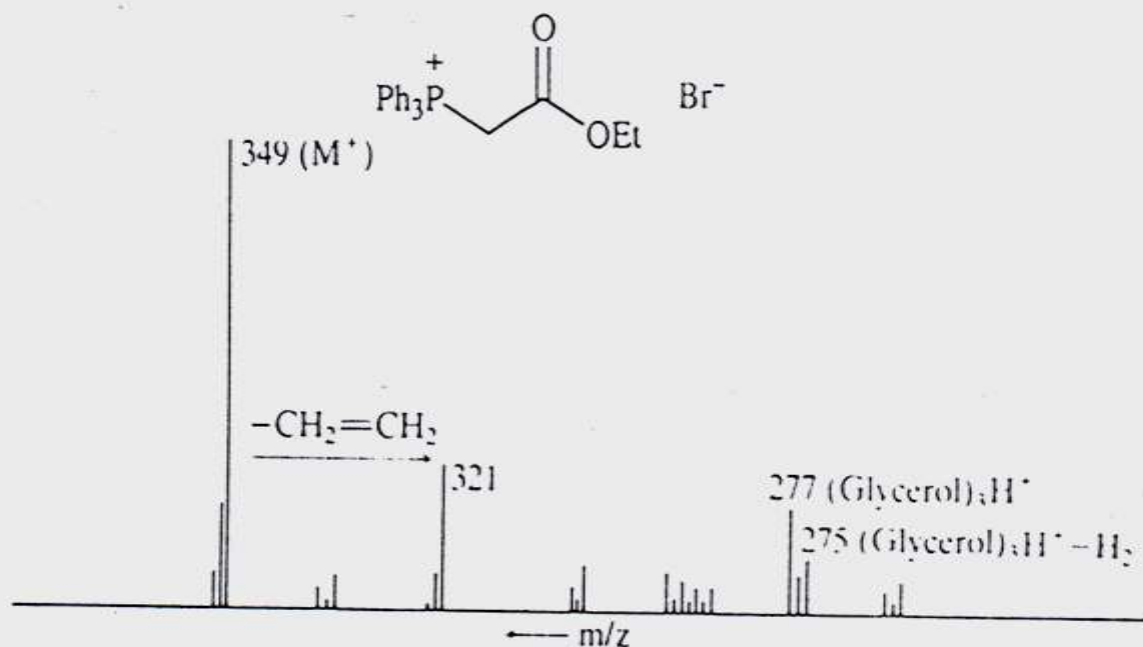
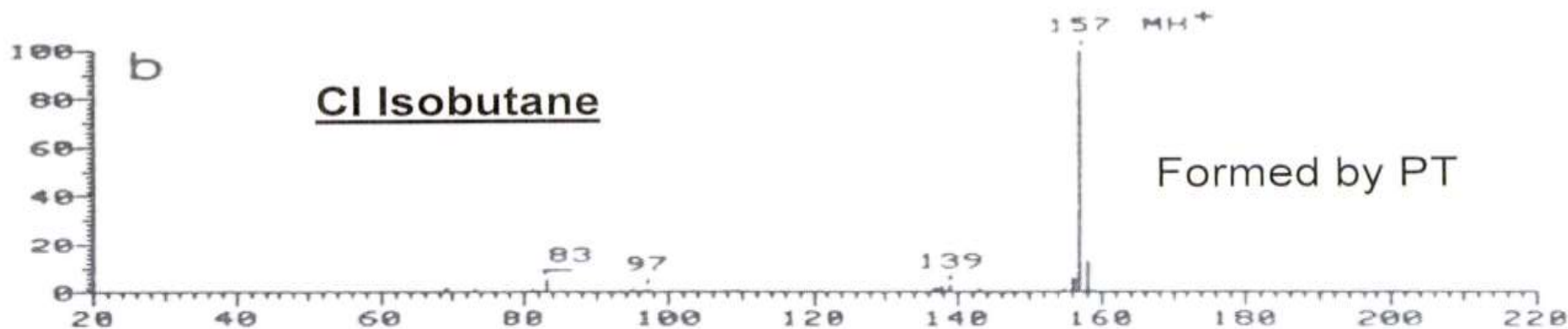
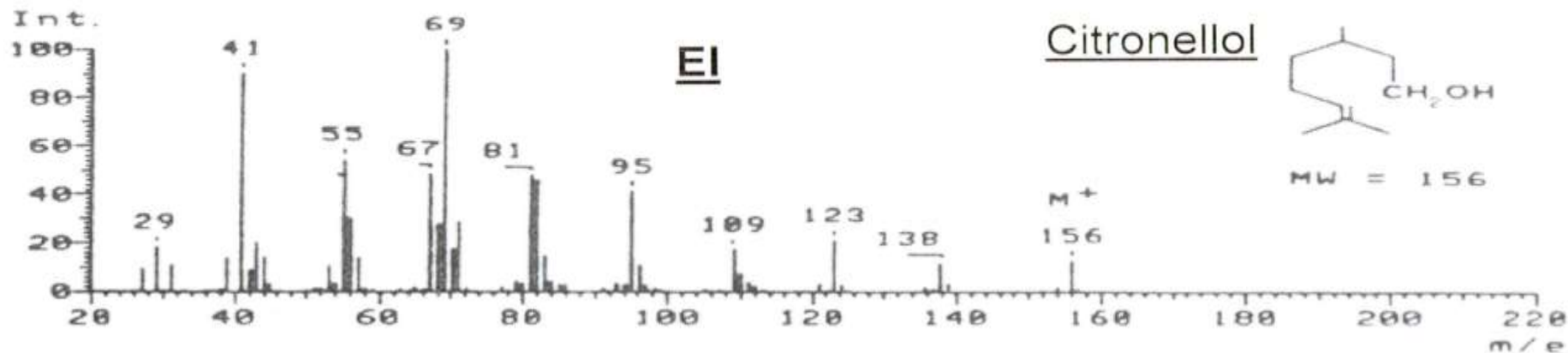


Fig. 4.31

Use CI to Simplify Spectra



(a) EI mass spectrum of citronellol; (b) CI (isobutane) mass spectrum of citronellol

Change reagent gas to CH_4 ? To NH_3 ?

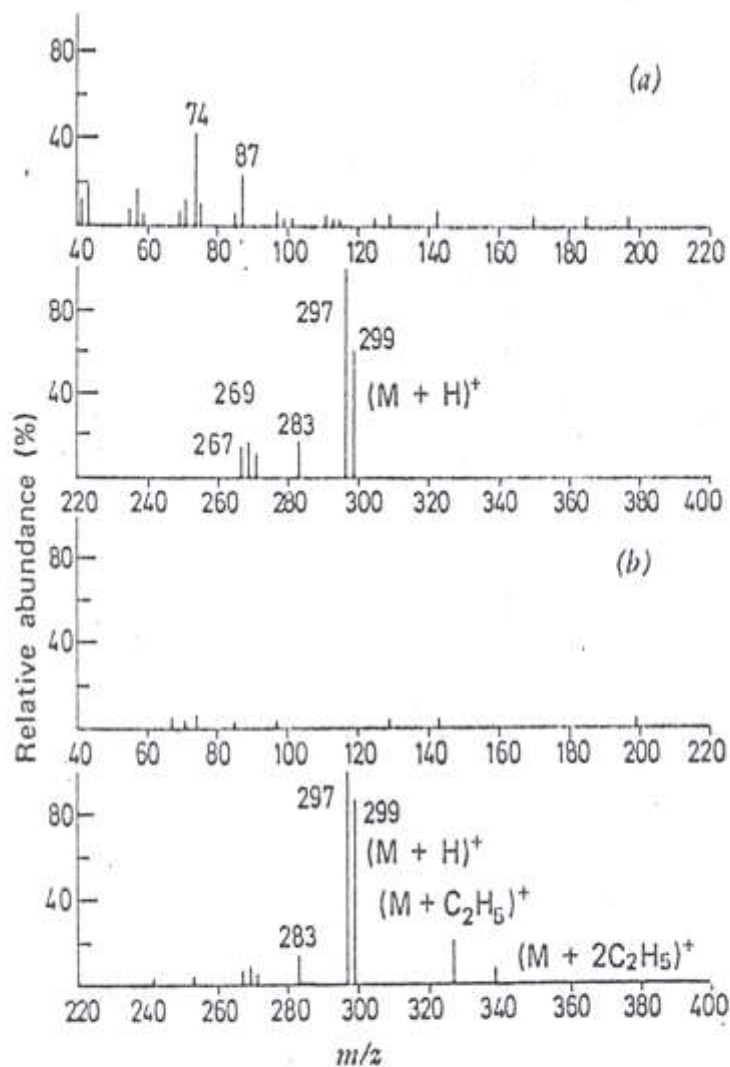


Figure 8-30. (a) CI (H_2) mass spectrum of methyl stearate. (b) CI (CH_4) mass spectrum of methyl stearate. DuPont 21-490 mass spectrometer equipped with high-pressure CI source. Source pressure about 0.5 torr, 200°C (71).

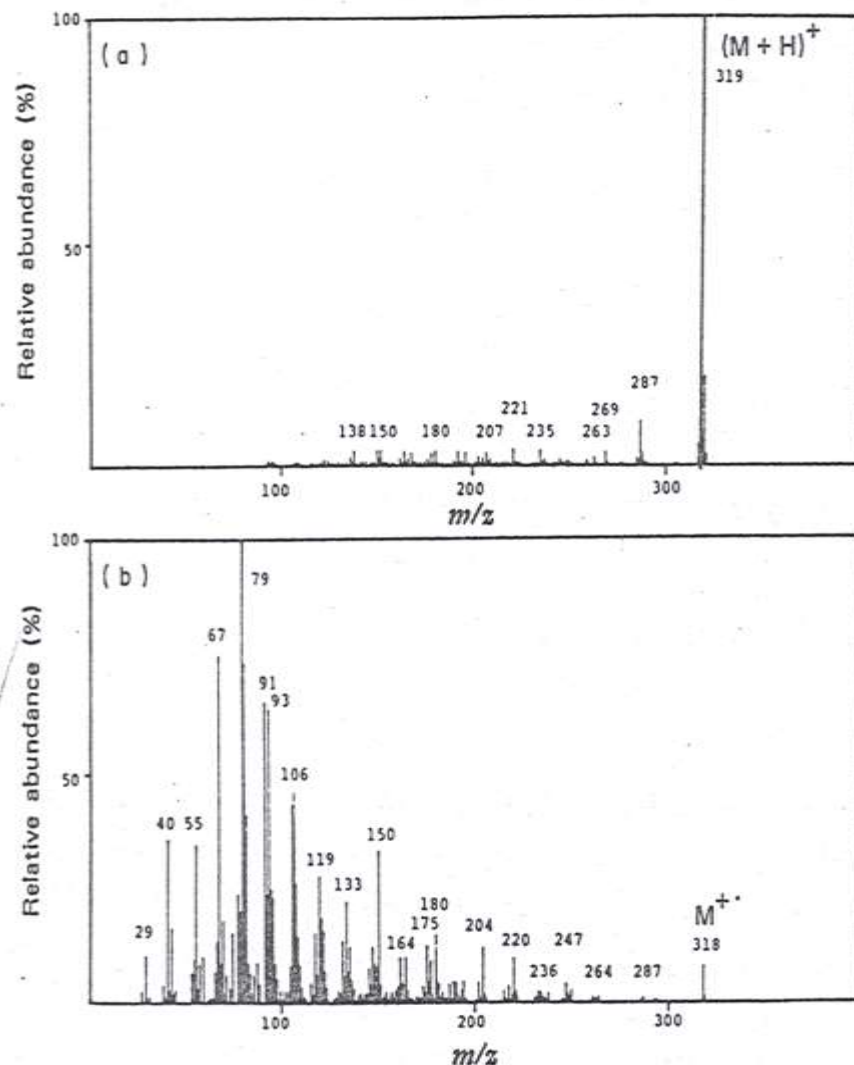


Figure 8-32. Mass spectra of methyl ester of $C_{20:4}$ acid. (a) Isobutane CI spectrum. Shimadzu LKB-9000 GC/MS. Ion source pressure 0.7 torr, 250°C. (b) EI spectrum with same instrument, 70 eV, 250°C (75).

The power of CI in determining molecular weights of volatile compounds is illustrated in Fig. 4.28. Although the EI mass spectrum of dioctyl phthalate contains no peak due to M^+ , abundant MH^+ peaks m/z 391 are observed in both the CH_4 and isobutane CI spectra. In accord with the considerations in Sec. 4.2, the isobutane spectrum contains less abundant fragment ions. The formation of the fragment ions is rationalized in Scheme 4.4.

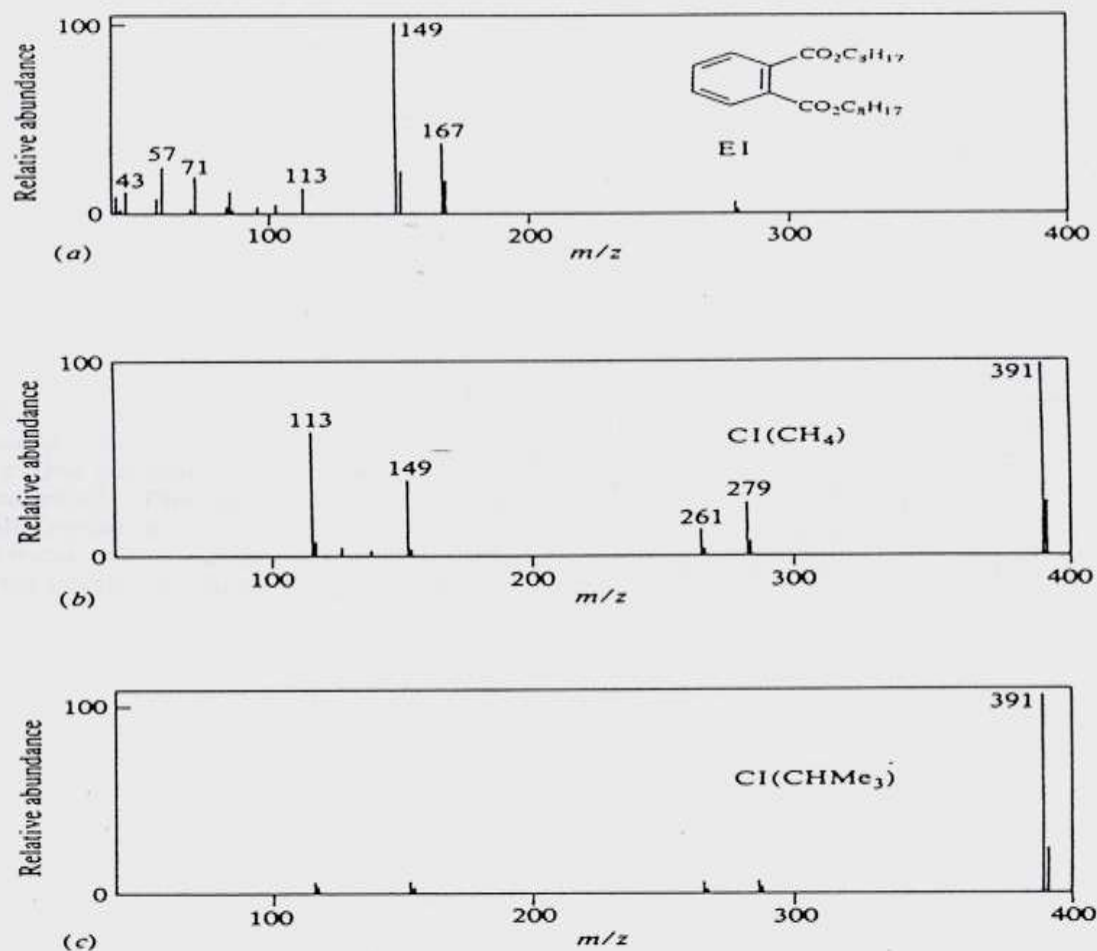


Fig. 4.28

(Reprinted with permission from Howe, Williams, and Bowen, 1981.)

Choice of reagent gas:

Two factors determine the choice of the gas to be used:

- Proton affinity (PA): $\text{H}^+ + \text{M} \rightarrow \text{MH}^+$
- (Energy transfer)
- Excitation energy of $\text{MH}^+ = \text{PA}(\text{M}) - \text{PA}(\text{R})$
- Excitation energy in MH^+ facilitates fragmentation

NH_3 (ammonia) is the most used reagent gas in CI because of the low energy transfer of NH_4^+ compare to CH_5^+ for example. With NH_3 as reagent gas, usually MH^+ and MNH_4^+ (17 mass units difference) are observed.

Table 1. Reagent species and their properties for the common PCI gases.

Reagent Gas	Major Species	CI Products	Proton Affinity (kcal/mole)
Methane	CH_5^+	M+1	131.6
	C_2H_5^+	M+29	162.6
	C_3H_5^+	M+41	—
Isobutane	C_4H_9^+	M+1	195.9
Ammonia	NH_4^+	M+1	204.0
		M+18	—

Analysis and Separation of Sample Ions

- The main function of the **mass analyser** is to **separate** , or **resolve** the ions formed in the ionisation source of the mass spectrometer according to their **mass-to-charge (m/z)** ratios. There are a number of mass analysers currently available, the better known of which include **quadrupoles** , **time-of-flight (TOF)** analysers, **magnetic sectors**.

SEPARATION OR ANALYSIS OF IONS

- Magnetic Sector Mass Spectrometers
- Quadrupole Mass Spectrometers
- Time-of-Flight Mass Analyzers
- Tandem (MS-MS) mass spectrometry

- The m/z scale of the mass spectrometer is **calibrated** with a known sample that can either be analysed independently (external calibration) or pre-mixed with the sample and matrix (internal calibration).

In a magnetic deflection mass spectrometer, ions leaving the ion source are accelerated to a high velocity. The ions then pass through a **magnetic sector** in which the magnetic field is applied in a direction perpendicular to the direction of ion motion. From physics, we know that when acceleration is applied perpendicular to the direction of motion of an object, the object's velocity remains constant, but the object travels in a circular path. Therefore, the magnetic sector follows an arc.

- A magnetic sector alone will separate ions according to their mass-to-charge ratio. However, the resolution will be limited by the fact that ions leaving the ion source do not all have exactly the same energy and therefore do not have exactly the same velocity. This is analogous to the *chromatic aberration* in optical spectroscopy. To achieve better resolution, it is necessary to add an **electric sector** that focuses ions according to their kinetic energy. Like the magnetic sector, the electric sector applies a force perpendicular to the direction of ion motion, and therefore has the form of an arc.

Electric Sector/Double Focusing Mass Spectrometers. The voltage is set so that the beam follows the curve of the analyzer. The radius of the ion trajectory (r) depends upon the kinetic energy of the ion (V) and the potential field (E) applied across the plates.

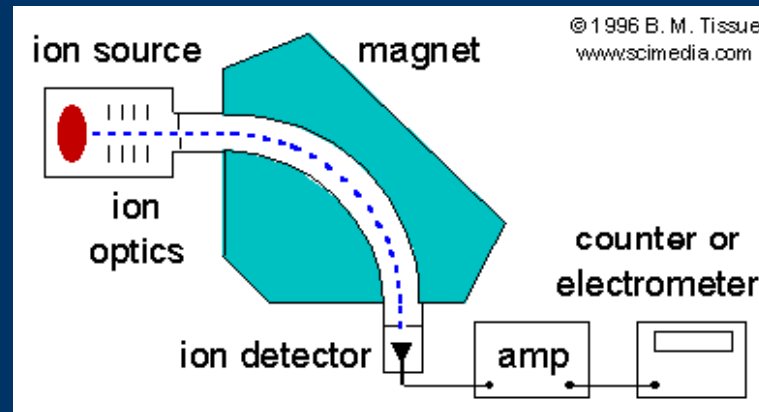
$$r = 2V/E$$

Equation shows that an electric sector will not separate ions accelerated to a uniform kinetic energy. The radius of the ion beam is independent of the ion's mass to charge ratio so the electric sector is not useful as a standalone mass analyzer.

The electric sector is a kinetic energy analyzer. In the source region of the mass spectrometer all ions are accelerated to the same kinetic energy. Because they have the same kinetic energy, they are not separated by an electric sector

An electric sector is, however, useful in series with a magnetic sector. An electric sector significantly improves the resolution of the magnetic sector by focussing the ions.

Magnetic-Sector Mass Spectrometry



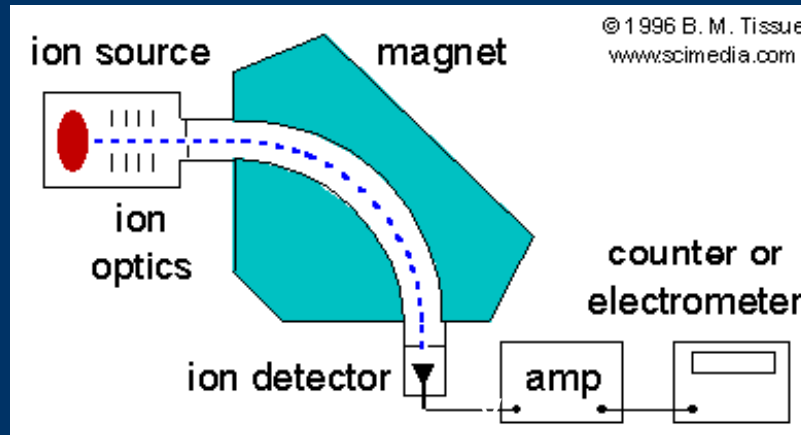
THEORY:

The ion source accelerates ions to a kinetic energy given by:

$$KE = \frac{1}{2} mv^2 = eV$$

Where m is the mass of the ion, v is its velocity, e is the charge on the ion, and V is the applied voltage.

Magnetic-Sector Mass Spectrometry



- The ions enter the flight tube and are deflected by the magnetic field, B .
- Only ions of mass-to-charge ratio that have equal *centripetal* and *centrifugal* forces pass through the flight tube:

$$mv^2 / r = Bev, \text{ where } r \text{ is the radius of curvature}$$

Magnetic-Sector Mass Spectrometry

$$mv^2 / r = Bev$$

By rearranging the equation and eliminating the velocity term using the previous equations,

$$m/e = B^2 r^2 / (2V)$$

This equation shows that the m/e ratio of the ions that reach the detector can be varied by changing either the magnetic field (B) or the applied voltage.

In summary, by varying the voltage or magnetic field of the magnetic-sector analyzer, the individual ion beams are separated spatially and each has a unique radius of curvature according to its mass/charge ratio.

The electric sector is usually held constant at a value which passes only ions having the specific kinetic energy, V . Therefore the parameter that is most commonly varied is B , the magnetic field strength.

An alternative is to hold B constant and scan V . The electric sector potential tracks the accelerating voltage, so the relationship between mass-to-charge ratio and accelerating voltage is a simple linear relationship.

The disadvantage of an accelerating voltage (electric field) scan is that the sensitivity is roughly proportional to the mass-to-charge ratio.

Time of Flight Mass Spectrometry (TOF-MS)

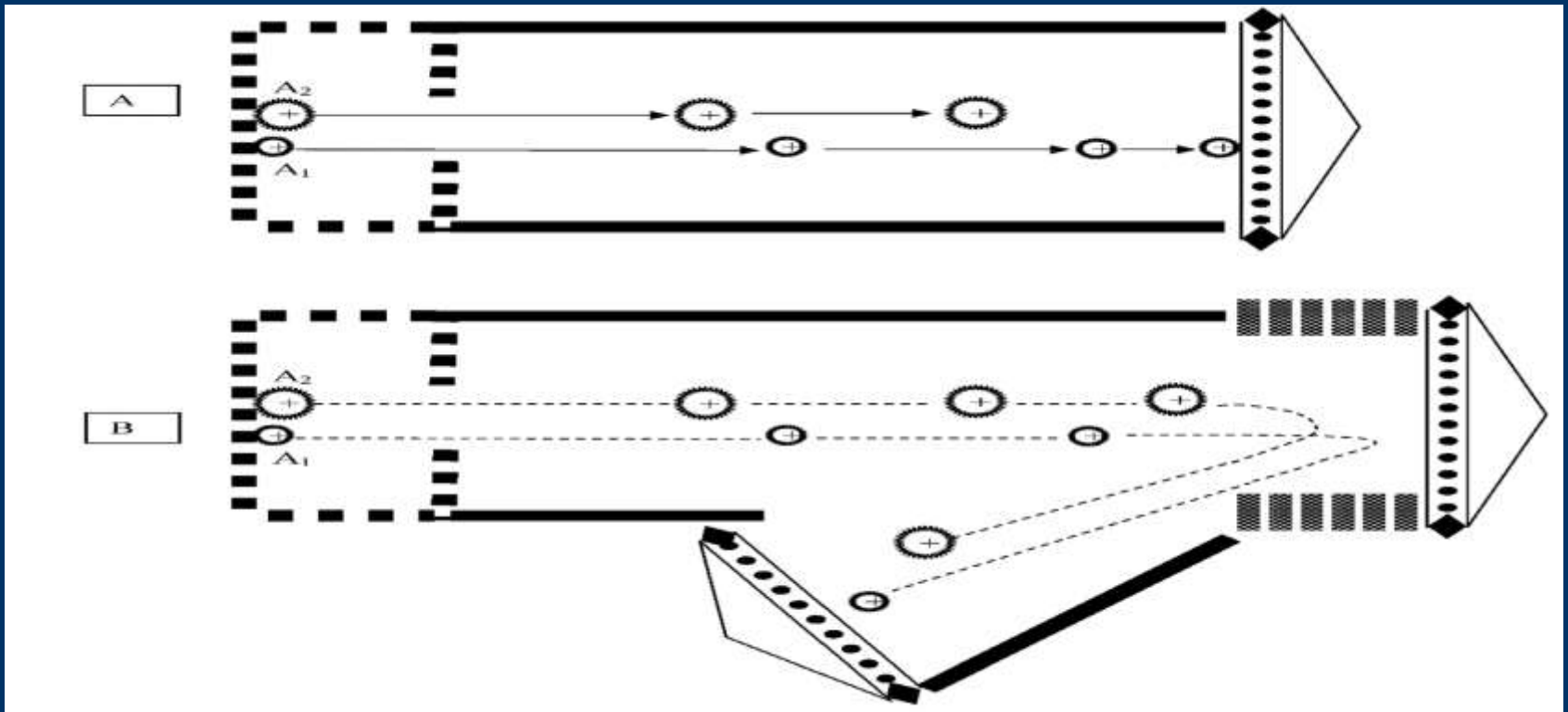
INTRODUCTION:

- Separates ions based on flight time
- Operates in pulsed mode
- Ions accelerated by an electric field
- Lighter ions reach the detector first

Time of Flight Mass Spectrometry (TOF-MS)

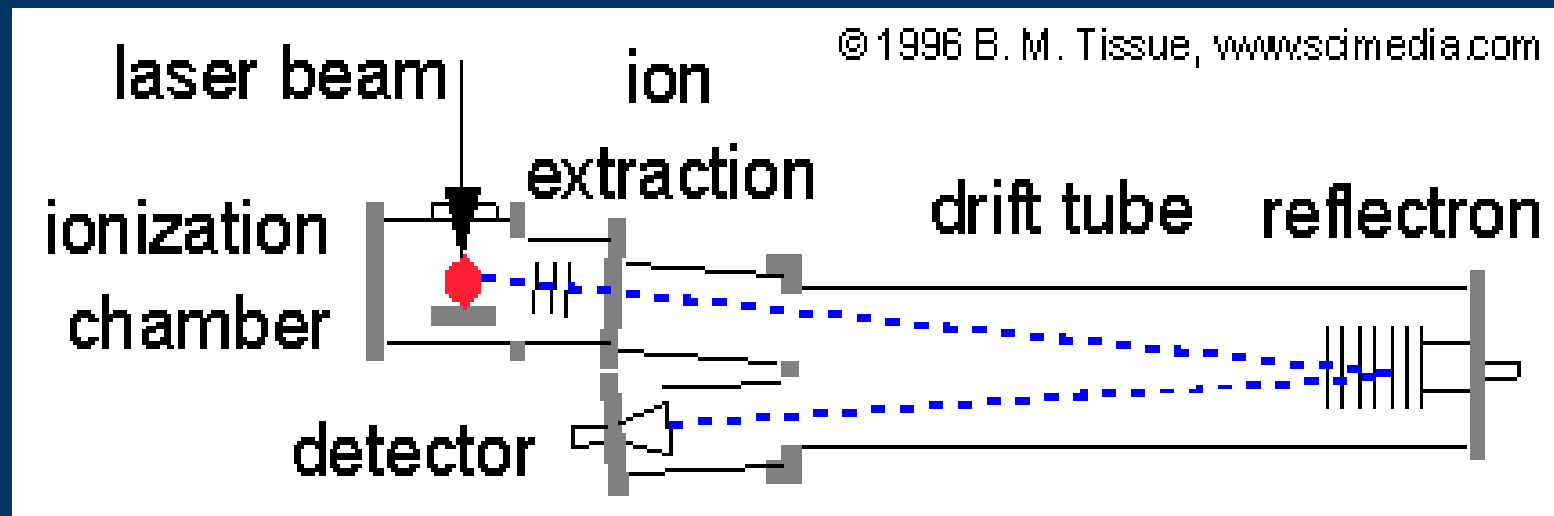
THEORY:

- $KE = eV$ when ions are accelerated through an electric field
- KE of ion is $\frac{1}{2}mv^2$, so $eV = \frac{1}{2}mv^2$ and velocity is inversely proportional to mass
- Transit time (t) is L/v , where L is drift tube length and v is velocity
- So $t = L/(\frac{2V}{m/e})^{1/2}$ can be solved for charge-mass ratio



Time of Flight Mass Spectrometry (TOF-MS)

HOW IT'S DONE:



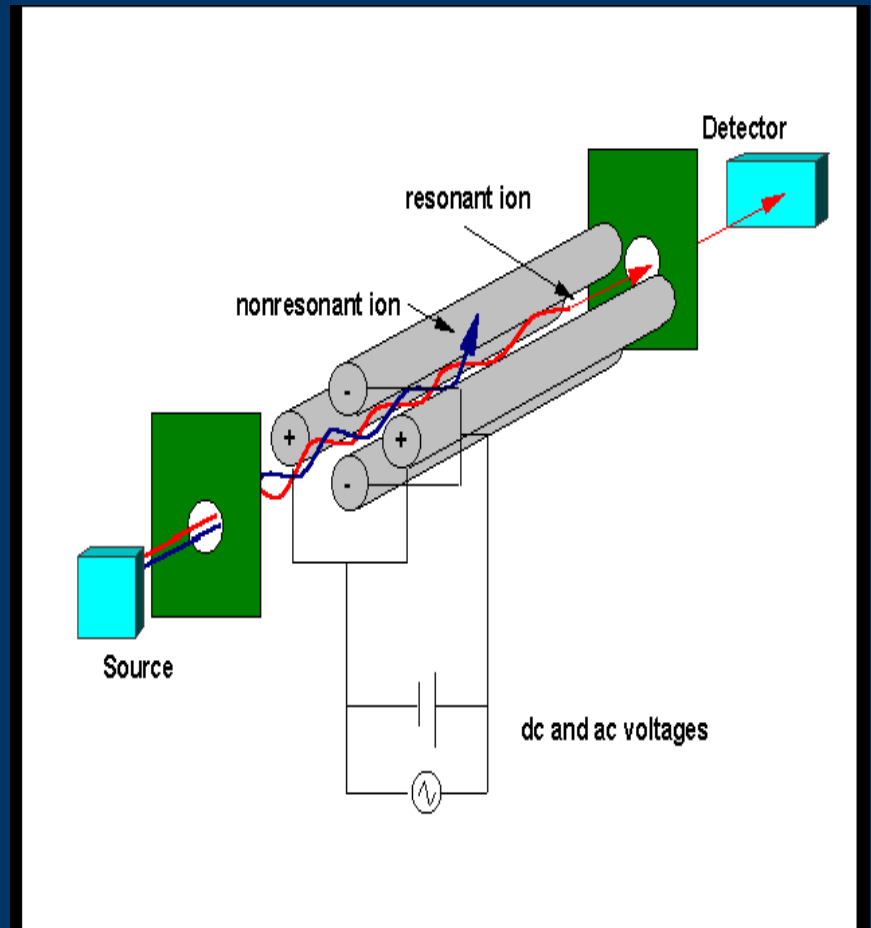
A reflectron is an ion optic device in which ions in a time-of-flight mass spectrometer pass through a "mirror" or "reflectron" and their flight is reversed.

Produces a mass spectrum as a function of time (can be measured every 10 nsec)

Though ToF has the advantage of being able to detect a very high mass range, the tube linearity of a conventional ToF analyzer (Figure 5a) can influence its resolving power, because ions entering the ToF carry different kinetic energies (KE) and this subsequently will affect resolution and molecular ion measurements. This drawback was overcome by the development of the reflectron (electrostatic ion mirror) (61), which is an ion optic device that changes the path of the ions within the ToF, as shown in Figure 5b. Ions with higher KE will penetrate deeper into the ion mirror and hence ions will be gradually repelled, improving the resolution of the ToF spectrum. Another factor is the tube length; because ions (in reflectron-ToF) are traveling a longer path, better resolution as well as accurate mass measurements can be obtained due to the increase in the flight time (62). Reflectron-type mass analyzers usually have two detectors: the first is beyond the ion mirror, and the second is at the end of the ion path. Due to the fact that a reflectron can result in ion loss, the operator can decide whether to compromise resolution or sensitivity. Due to the nature of ToF, it is widely linked to the MALDI ion source (i.e., MALDI-ToF). Nevertheless, ToF

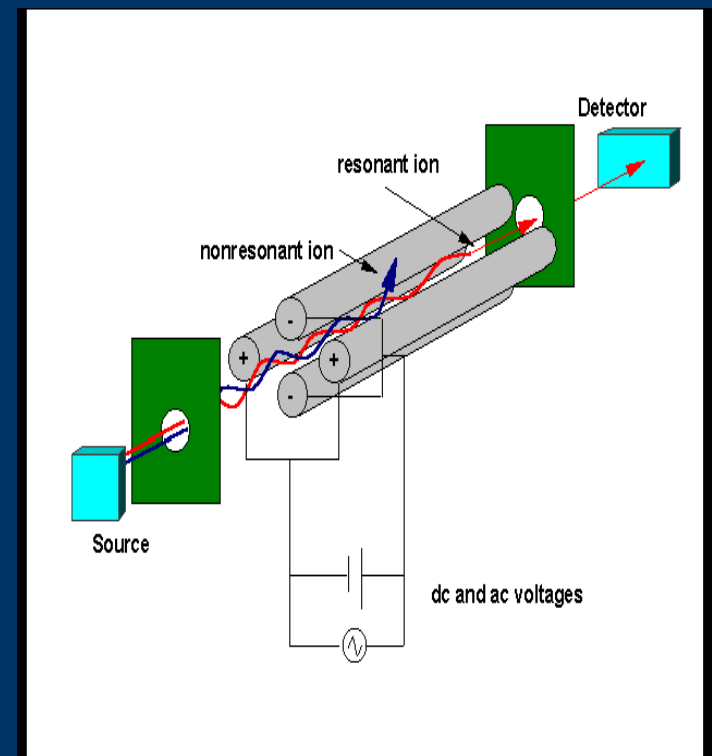
Basis of Quadrupole Mass Filter

- consists of 4 parallel metal rods, or electrodes
- opposite electrodes have potentials of the same sign
- one set of opposite electrodes has applied potential of $[U+V\cos(\omega t)]$
- other set has potential of $-[U+V\cos\omega t]$
- U = DC voltage, V = AC voltage, ω = angular velocity of alternating voltage to a stable vs. unstable ion trajectory through the quadrupole rods.



Operation of Quadrupole Mass Filter

- voltages applied to electrodes affect trajectory of ions with the m/e ratio of interest as they travel down the center of the four rods
- these ions pass through the electrode system
- ions with other m/z ratios are thrown out of their original path
- these ions are filtered out or lost to the walls of the quadrupole, and then ejected as waste by a vacuum system
- in this manner the ions of interest are separated



Obtaining A Spectrum

Quadrupole Mass Analyzers

- a mass spectrum is obtained by varying the voltages on the rods and monitoring which ions pass through the filter
- two methods for varying rod voltages:
 - vary ω while holding U and V constant
 - vary U and V but keep the ratio U/V fixed
 - remember: U= DC voltage, V=AC voltage, ω = angular velocity of alternating voltage
- The resolution is determined by the magnitude of U/V ratio
- beam currents for individual m/e ratios may be as low as 10^{-13} A and so electron multipliers are usually used

Direct current (DC) is an electrical current that is unidirectional as a result of a voltage source whose output maintains the same polarity. An example is the output of a flashlight battery. DC is not RF nor is it part of the electromagnetic spectrum.

Alternating current (AC) is an electrical current that periodically reverses or alternates in direction as a result of a voltage source whose output periodically reverses or alternates in polarity.

The general principle of operation of the mass filter can be visualized *qualitatively* as follows:

light ions (low mass to charge ratio) are able to follow the alternating component of the field. For the X-direction, those ions will stay in phase with the RF drive, gain energy from the field and oscillate with increasingly large amplitude until they encounter one of the rods and are discharged. Therefore the X-direction is a **high-pass mass filter**:

only high masses will be transmitted to the other end of the quadrupole without striking the X-electrodes.

On the other hand, in the Y-direction, heavy ions will be unstable because of the defocusing effect of the DC component, but some lighter ions will be stabilized by the AC component if its magnitude is such as to correct the trajectory whenever its amplitude tends to increase. Thus the Y-direction is a **low-pass mass filter**:

only low masses will be transmitted to the other end of the quadrupole without striking the Y electrodes. By a suitable choice of RF/DC ratio, the two directions together give a mass filter which is capable of resolving individual atomic masses.

Background Info

Quadrupole Mass Analyzers

- are in use since the 1950s
- most commonly used mass spec today
- sometimes referred to as “mass filters” because ions of only a single mass to charge (m/z) ratio pass through the apparatus
- separate ions based on oscillations in an electric field (the quadrupole field) using AC and DC currents

Benefits

Quadrupole Mass Analyzers

- easy to use
- simple construction
- fast
- low cost
- can achieve unit to 0.1 m/q resolution

- tolerant of relatively poor vacuums ($\sim 5 \times 10^{-5}$ torr), which make them well suited to electrospray ionization (because these ions are produced under atmospheric conditions)
- quadrupoles are now capable of routinely analyzing up to a m/z ratio of 3000, which is useful in electrospray ionization of biomolecules, which commonly produces a charge distribution below m/z 3000

Summary

- Magnetic Mass Spectroscopy shows the relationship between voltage and mass to charge ratio:

$$m/e = B^2 r^2 / (2V)$$

- Time of Flight Mass Spectroscopy shows the relationship between tube time and mass to charge ratio:

$$m/e = 2Vt^2 / L^2$$

- Electric Quadrupole shows the relationship between AC/DC currents and mass to ratios.

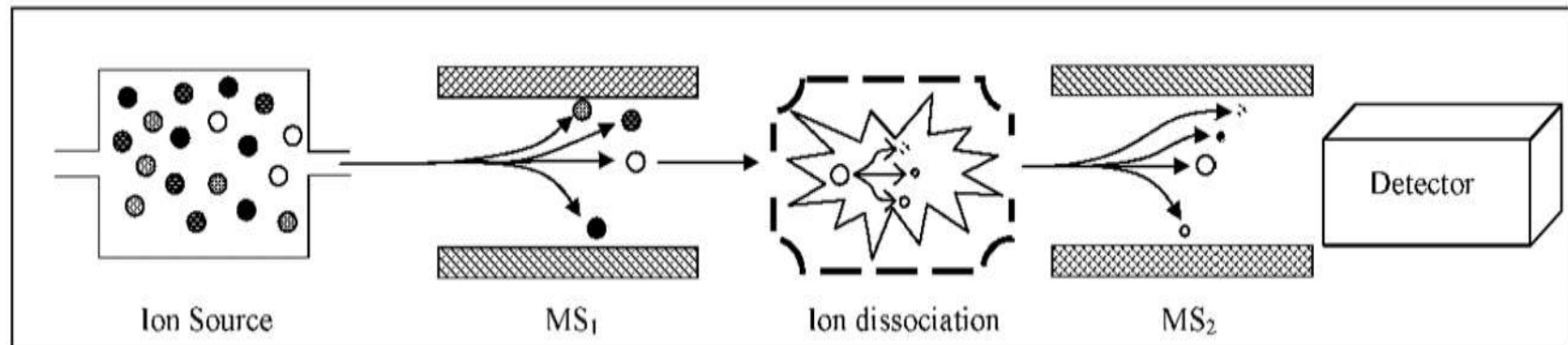
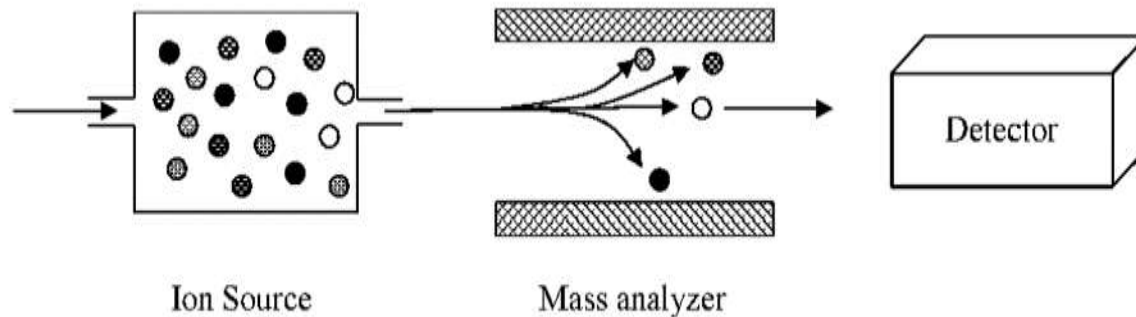
Tandem (MS-MS) mass spectrometry

- Tandem (MS-MS) mass spectrometers are instruments that have more than one analyser and so can be used for structural and sequencing studies. Two, three and four analysers have all been incorporated into commercially available tandem instruments, and the analysers do not necessarily have to be of the same type, in which case the instrument is a **hybrid** one. More popular tandem mass spectrometers include those of the **quadrupole-quadrupole**, **magnetic sector-quadrupole** and more recently, the **quadrupole-time-of-flight** geometries.

Tandem (MS-MS) mass spectrometry

- **Tandem mass spectrometry (MS-MS)** is used to produce **structural information** about a compound by fragmenting specific sample ions inside the mass spectrometer and identifying the resulting fragment ions. This information can then be pieced together to generate structural information regarding the intact molecule. Tandem mass spectrometry also enables specific compounds to be detected in complex mixtures on account of their specific and characteristic fragmentation patterns.

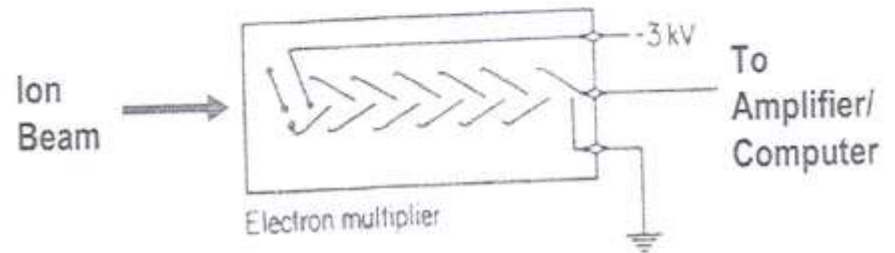
Tandem mass spectrometry, also known as **MS/MS** or **MS²**, involves multiple steps of mass spectroscopy selection, with some form of fragmentation occurring in between the stages. In a tandem mass spectrometer, ions are formed in the ion source and separated by mass-to-charge ratio in the first stage of mass spectrometry (MS₁). Ions of a particular mass-to-charge ratio (precursor ions) are selected and fragment ions (product ions) are created. The resulting ions are then separated and detected in a second stage of mass spectrometry (MS₂).



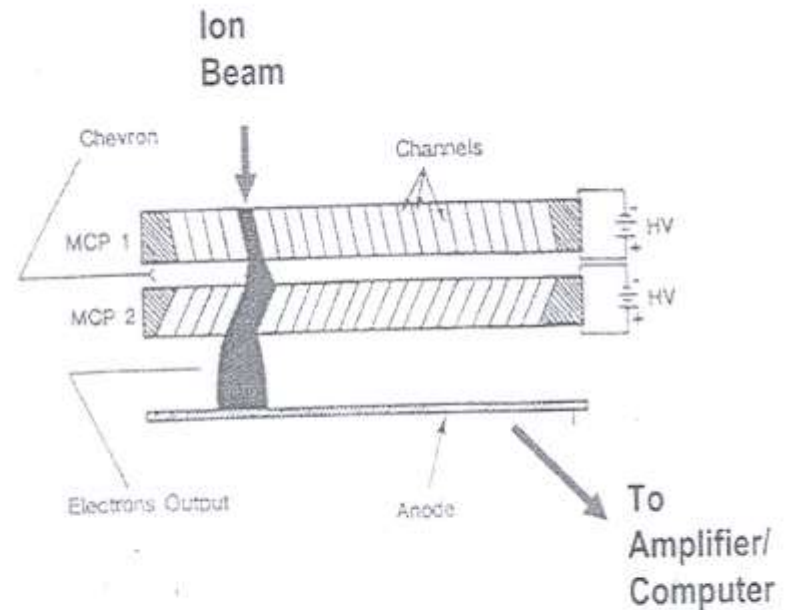
Steps in Mass Spectrometric Analyses

Ion Detection and Recording

Electron Multiplier: Converts each ion into a packet of 10^6 electrons. Electrical current from electrons is recorded by a computer. Used with magnetic sector, quadrupole, and ion trap instruments



Array Detector: Joins many small electron multipliers into one unit. Used with TOF to increase area of detection (sensitivity)



SUMMARY

- Electrospray Ionization (ESI)
- ESI Versus EI (Advantages)
- ESI Versus EI (Difficulties)
- Applications
- MALDI
- Negative and Positive Ions
- Analyzers
- Magnetic Sector Analyzers
- Benefits of Magnetic Sector Analyzers
- Limitations of Magnetic Sector Analyzers
- Quadrupole Analyzers
- Time of Flight Analyzers
- Tandem MS-MS
- Detectors

Table 1. Summary of the various features of current ionization methods used in mass spectrometry

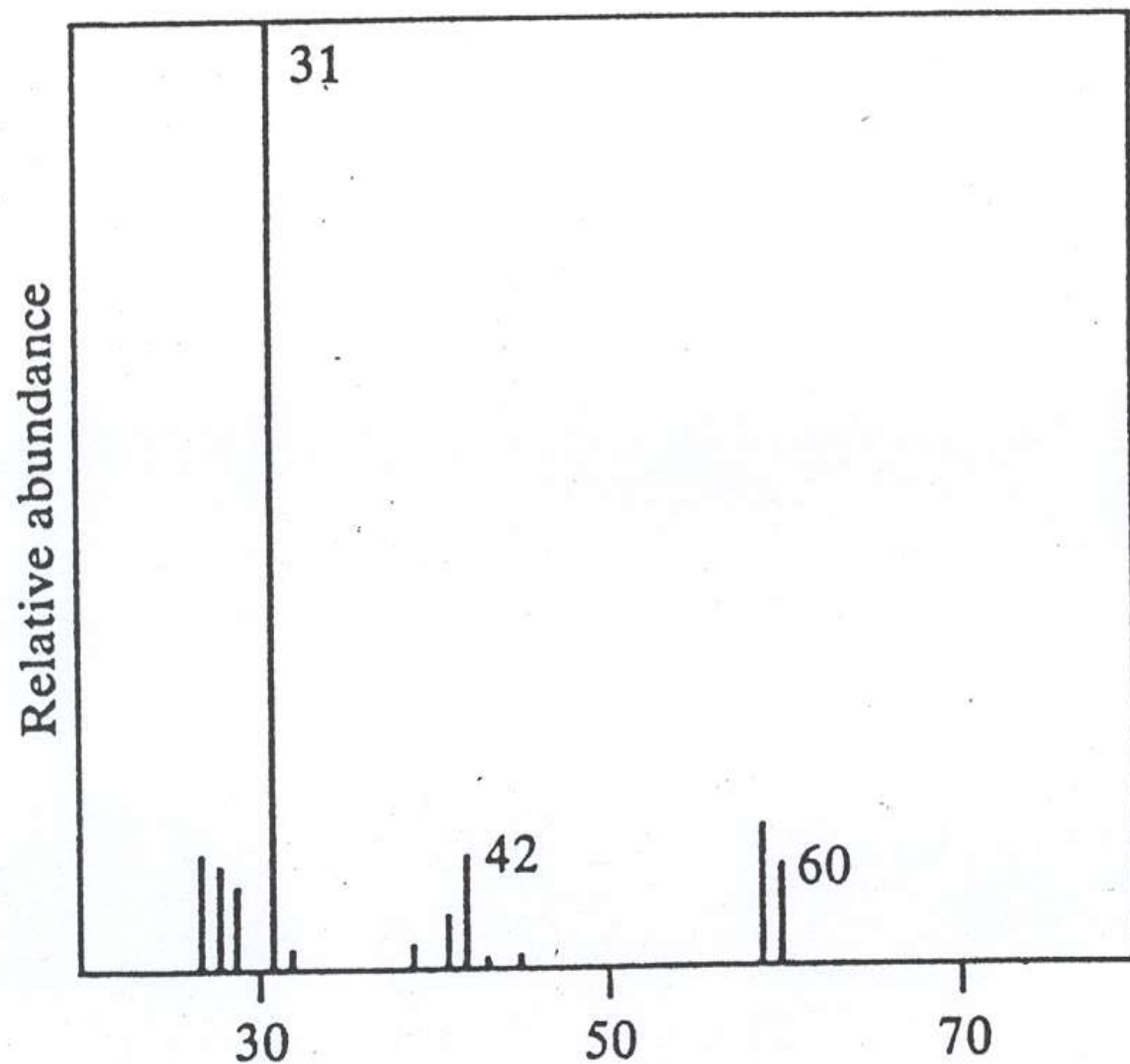
Ionization technique	Nature of analytes	Sample introduction	Mass range	Brief description
Electron impact (EI)	Volatile; thermally stable	Gas chromatography solid or liquid probe	<1000 Da	Hard method; mainly fragment ions
Chemical ionization (CI)	Volatile; thermally stable	Gas chromatography solid or liquid probe	<1000 Da	Soft method; Molecular ion
Fast atom bombardment FAB	Organometallic compounds	Liquid chromatography, direct injection	<5000 Da; optimal range 200–2000	Soft method; require matrix
Matrix-assisted laser desorption ionization (MALDI)	Biomolecules (proteins, DNA, glycoconjugates)	Sample is cocrystallized with a matrix	Can go beyond 500,000 Da	Very soft method singly charged ions
Electrospray ionization (ESI)	Organic and inorganic compounds	Sample in solution	From very low masses to extraordinarily high	Very soft method; multiply charged ions

EXAMINATION OF THE MASS SPECTRUM

A. *Recognition of the Molecular ion*

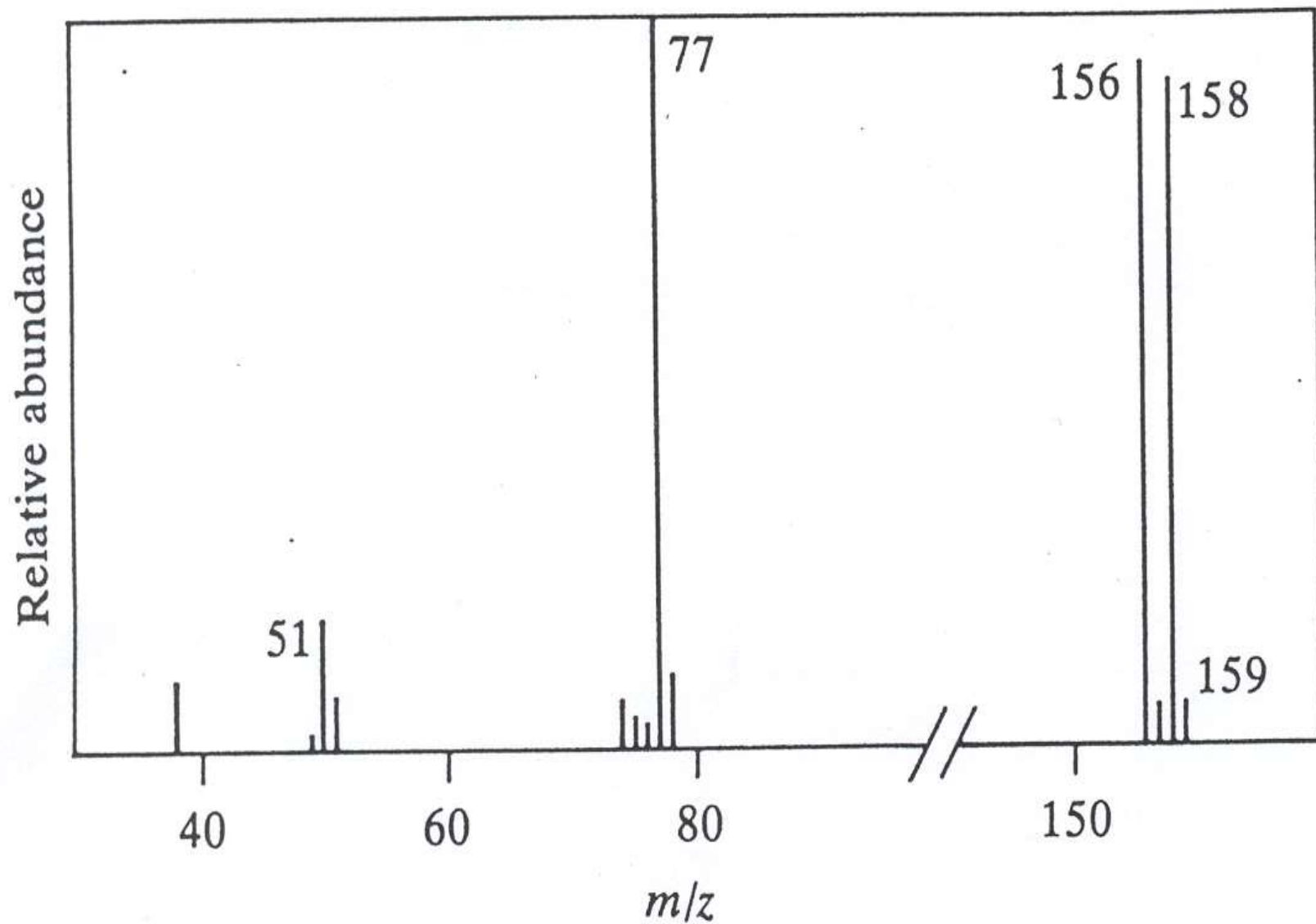
- a) Often $(M-1)^+$ and sometime $(M-2)^+$, $(M-3)^+$ etc. are observed due to losses of H atoms from M^+ .

Figure 11.2. Normalized line diagram taken from mass spectrum of compound A in figure 11.1.



b) It is accompanied by isotope ions. For Cl, Br, S, $M+2$ are more abundant compared with the $(M+2)^+$ ions of similar compounds without these heteroatoms. [Isotope abundance].

Figure 11.3. Mass spectrum of compound B. Ratio of peaks m/z 159: m/z 158 = 6.2:92.5.

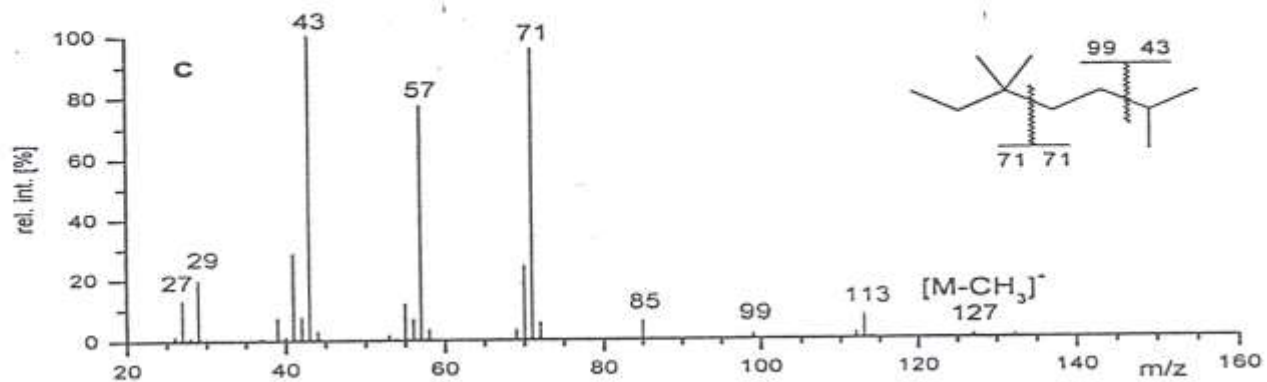
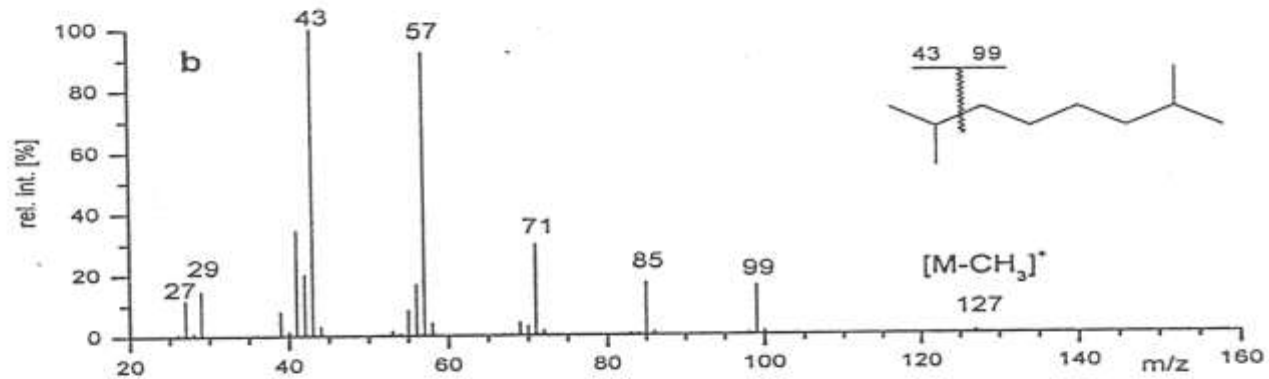
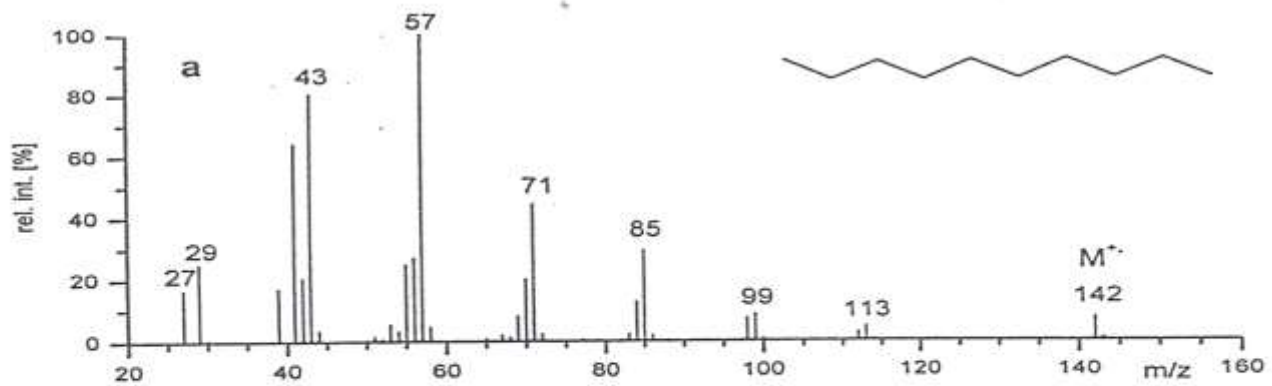


c) In nitrogen compounds most abundant ion is $(M-1)^+$ and not the M^+

In case, (a) and (c) it is worthwhile examining the spectrum by a soft ionization technique.

d) Absence of molecular ions (or extremely weak M^+) suggests highly branched molecules. [Soft ionization techniques]

- e) Organic compounds usually have C,H,N,O,S, Cl, Br, etc. and it is very unlikely that M^+ give rise to losses of 5-14 and 21-25 units. If M^+ shows such losses it is most unlikely to be correctly assigned.
- f) losses of H & CH_3 (1 & 15 mass units) are commonly observed.
- g) M^+ ions must contain all the elements that are evident elsewhere in the spectrum.



B)

- If metastable ions which implicate the candidate peak as being due to product ions of some higher mass species, then it can not be the true M^+ peak.
- Metastable ions in the routine spectrum; recall that the position of the metastable ion (m) is given by,

$$m = \frac{m_2^2}{m_1}$$

where,

m_1 = mass of the precursor

m_2 = mass of the product

C) It is useful to remember that:

- i) An odd molecular weight indicates an odd No. of N atoms.
- ii) Abundant $(M+2)^+$ ions would suggest the presence of Cl or Br and less abundant ones the presence of S.
- iii) An approximate No. of C atoms can be gained from the relative heights of M^+ (h) and $(M+1)^+$ (h') peaks since ^{13}C is present in 1.1% natural abundance in organic compounds., the approx. No. of C atoms is given by $100 h'/1.1h$
(e.g. ms of naphthalene $\text{C}_{10}\text{H}_{10}$:

$$\frac{100 \times 10.9}{1.1 \times 100} = \sim 10$$

- D) to obtain the mol.formula — it is necessary to measure the accurate mass of M^+ ions and obtain molecular formula from table or by calculation.
- E) The molecular formula obtained may then be used to calculate **the number of double bond equivalents**.
- F) The **general appearance** of the spectrum.
- i) A spectrum exhibiting many frag.ions increasing in abundance towards low m/z values indicates a predominantly **aliphatic structure**.
- ii) A spectrum with a few frag.ions, abundant mol.ions & doubly charged ions will suggest an **aromatic structure**.

- G) Next step is to note the *major frag.* ions and attempt to elucidate the main fragmentation pathways.
- H) In the absence of N atoms, frag.ions occurring at odd m/z values are even-electron species resulting from ***simple bond fission***; frag.ions at even m/z values are odd electron species produced by ***multiple bond cleavage*** suggesting that ***rearrangement*** may have occurred.

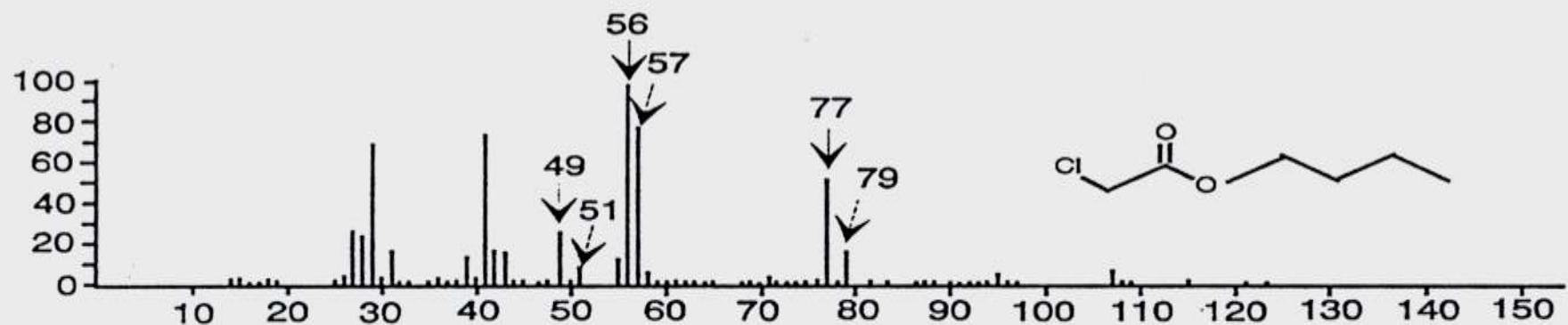
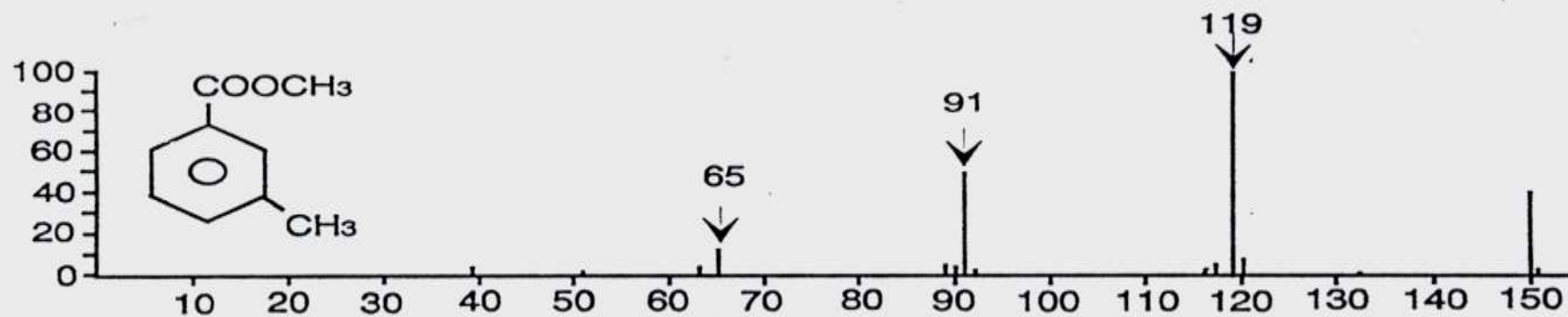
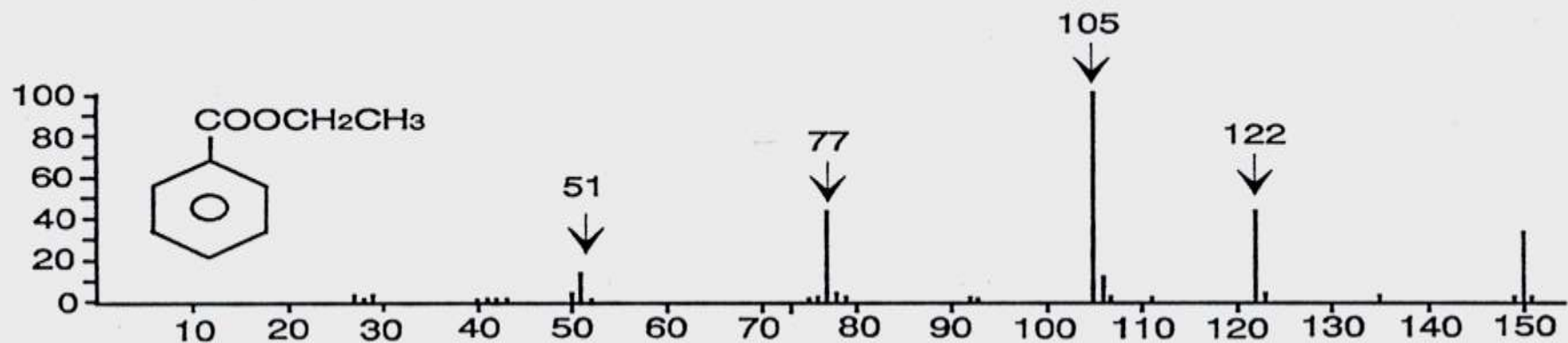


Figure 6.17: Spectra of three compounds having the same mass (150 Da) illustrating a few interpretative rules.

STRUCTURE ELUCIDATION

Without rationalization

Mass spectra of unknown structure may be compared with the published spectra compilations either:

- i) Manually $\longrightarrow$ tedious ; or
- ii) Through a computer and automatic library searching

Possible only if the mass spectrum has already been recorded and available in DATA.

By rationalization ——— (Most in use):

i) Functional-group approach

$M+18$ (loss of water)

$M+60$ (loss of $C_2H_4O_2$)

$M+31$ (loss of OCH_3 or $-CH_2OH$)

(Lists are available in text books)

ii) Accurate mass measurements

For example m/z 27 = HCN or C_2H_3 can be determined by obtaining exact mass of the ion at m/z 27.

iii) Characteristic m/z values

Peaks characteristic of a certain feature of molecular structure
e.g. m/z 43 (CH_3CO^+ , $C_3H_7^+$)

91 ($C_7H_7^+$)

m/z 105 C_6H_5CO (OR ??)

SUMMARY

- How to recognize the molecular ion.
(isotope peaks, rel.abundance, $M+H$ peaks, soft ionization for confirmation.)
- Examination of mass spectrum
(Aromatic, aliphatic, straight chain or branched)
- Odd molecular ion indicates odd no. of N.
- Look at major fragments
(Odd m/z values, in absence of N are even electron species -----simple bond fission)
(Even m/z values are odd electron species -----multiple bond cleavages.)
- Irrational approach-----data search.
- Rational approach.
- Use of accurate mass to differentiate b/w two or more elemental compositions.
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